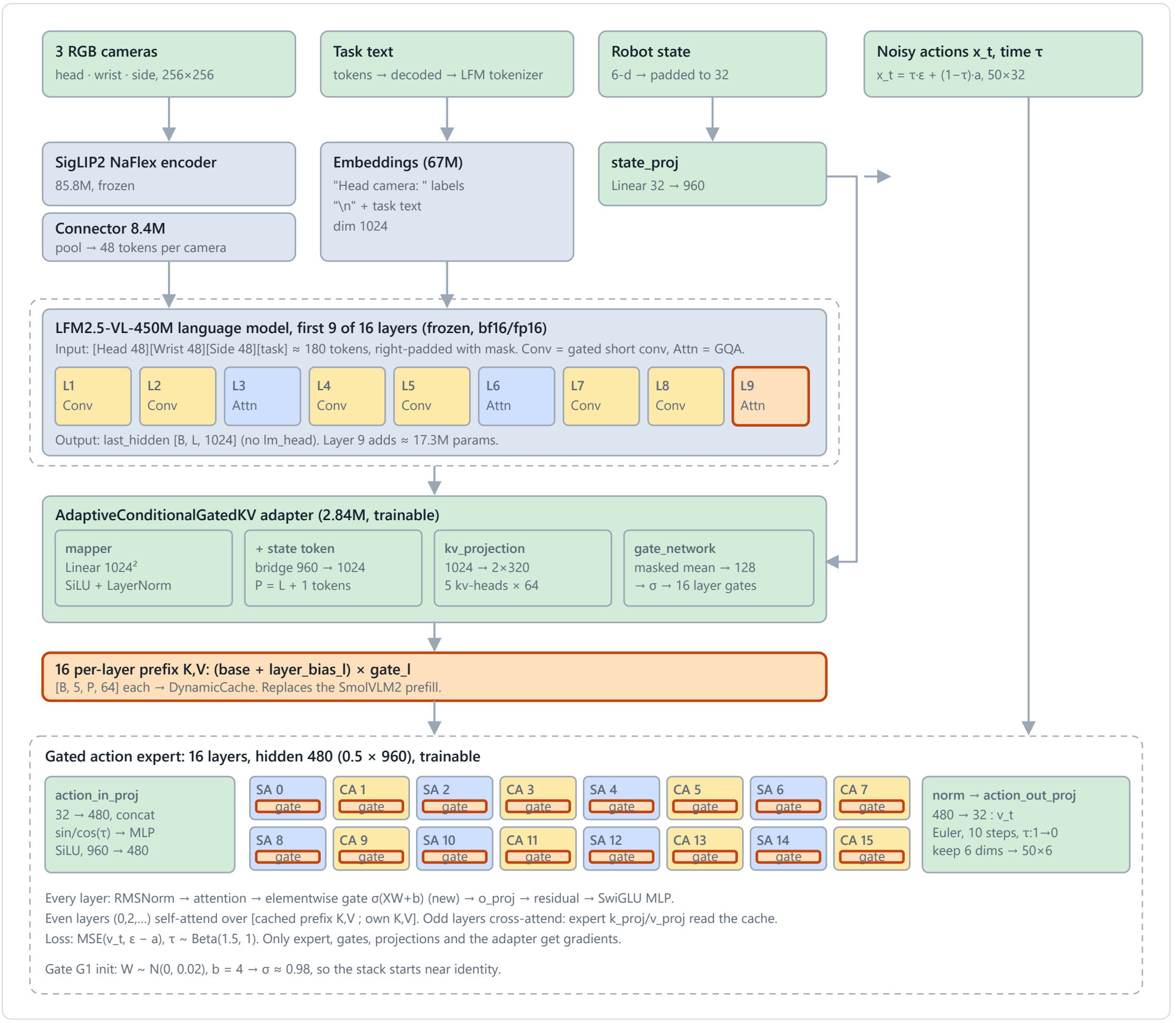
(adapter, gates, expert, flow-matching loss) is as in the notebook. Numbers use the training command (expert width 0.5, 3 cameras, 6-d state, chunk 50).

1. Full architecture

☐ Frozen ☐ Trainable ☐ New vs original SmolVLA / changed by the 9th layer ☐ Expert self-attention layer ☐ Expert cross-attention layer



2. Zoom: one expert layer with the gate

x (action tokens) → RMSNorm → Q from x → K, V: own (self) or from cache (cross) → SDPA, GQA 15q/5kv → $Y' = Y \odot \sigma(X_n \cdot W + b)$ → o_proj → + x → RMSNorm → SwiGLU → + residual

The gate reads the normed expert token, so it is query-dependent and can switch off individual channels per action step. It adds 16 × (480×960 + 960) ≈ 7.39M params at width 0.5.

3. Side by side with original SmolVLA

Original SmolVLA (smolvla_base)

Cameras 512×512 → SigLIP + pixel shuffle
64 tokens per camera

SmolVLM2-500M text model, first 16 layers
hidden 960, real prefill over images + text + state token

This policy (LFM2.5-VL 9L + gates)

Cameras 256×256 → SigLIP2 NaFlex + connector
pooled to 48 tokens per camera

LFM2.5-VL LM, first 9 of 16 layers (6 conv, 3 attn)
hidden 1024, text only; state is not fed to it

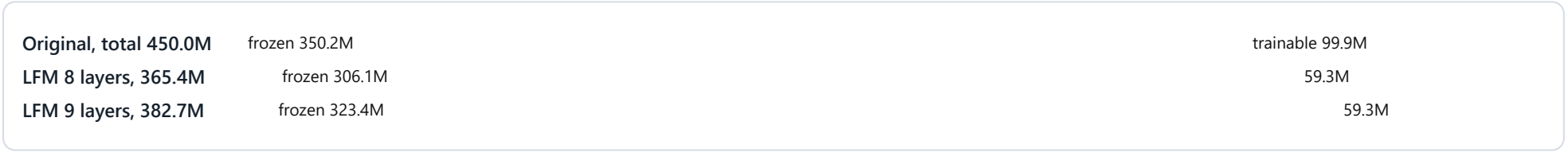
Real K,V cache from the VLM 16 layers, one K,V per layer	Trainable adapter synthesizes the K,V cache state token, per-layer bias and gate; 2.84M params
Expert 16 layers, hidden 720 (0.75×) SA/CA alternating, no gate	Expert 16 layers, hidden 480 (0.5×), 16 elementwise gates 136 checkpoint tensors skipped, fresh init
Flow matching, 10 Euler steps → chunk of 50	Flow matching, 10 Euler steps → chunk of 50 (unchanged)

> Key conceptual change: in SmolVLA the expert reads genuine VLM keys and values. Here the LFM hidden states are re-projected by a learned adapter, so the expert reads synthetic K,V. The 9th layer only deepens the frozen feature extractor; the expert still has 16 layers.

4. Numbers

Item	Original SmolVLA	This policy (9 layers)
Backbone	SmolVLM2-500M	LFM2.5-VL-450M
Backbone layers used	16 of 32	9 of 16 (8 in the notebook)
Backbone hidden size	960	1024
Visual tokens / camera	64	48
Prefix K,V source	VLM prefill	Gated-KV adapter
Expert layers / width	16 / 720	16 / 480
Attention gates	none	16, elementwise
Total params	450.0M	≈ 382.7M (365.4M with 8 layers + 17.3M)
Trainable params	99.9M (width 0.75)	59.3M
Frozen params	350.2M	≈ 323.4M
Inference GFLOPs, batch 1	826.6 (no gates)	≈ 824 (817.7 measured at 8L, width 0.75, plus ≈ 6 for layer 9; estimate)

Layer 9 is probably a full-attention layer, going by the LFM2 hybrid layout (check `layer_types[8]`). Its parameter count is 17.3M if so, or about 18.4M if it is a conv layer. The FLOPs delta is my estimate from ≈ 180 LM tokens; the notebook only measured 8 layers.



Caveat on fairness: the trainable counts differ partly because the expert is narrower (0.5× vs 0.75×), not only because of the backbone. At width 0.5 the pretrained expert weights no longer fit, so the notebook skips them and trains those tensors from scratch.